Amazon Developer Hackathon · Ring track · October 2026

Call First

Courier scams end at the front door.
So does this.

The problem

The one kind of elder fraud that has to come to the house.

The call, the fear and the secrecy all happen at a distance. **The handover does not.** A courier arrives for the cash or the gold, and that is the only moment anyone outside the conversation can see.

1,737
doorstep pickups counted by the FBI, 2023 to May 2025

$186M
taken in those pickups

98%
of the losses reported by people over 60, per the Boston field office

Why the obvious approach fails

Alerting on strangers is useless here. She is expecting the courier.

Every doorbell on the market can tell her someone unknown is at the door. She already knows. She was told he would come, by someone she believes.

- Call First watches for something else: **a visitor nobody declared in advance**, and **the door opening for them**.
- Visits and deliveries are declared calmly, beforehand — a carer on Tuesdays, a parcel this afternoon. Those ring in silence.
- Everything else is a question worth one sentence at the door.

How it works · one visit, end to end

An unexpected visitor rings

> “Before you hand anything over, call your family first, and ask for your family word.”

Played by the Ring chime, in her own hallway. The family word is the one code the scammer cannot know.

The door opens anyway

Sarah is told at once — the doorbell’s still image, how sure the system is (a door sensor saw it, or only how long he stayed), and one instruction: **call Margaret now**. A printable report sheet for IC3 and the FTC comes with it.

For the next seven days

The family is told as soon as anyone unexpected appears — **before** the door opens. Couriers come back.

The threat the design names first

The victim is on the scammer’s side, for now.

A protection the scammer can talk her out of is not a protection. So **every way to weaken it is loud**.

- Pausing, switching off, adding a visit or a delivery for today, adding a new trusted person: **each one tells the family first**.
- A pause **ends by itself** after twenty-four hours. Nothing stays off because someone forgot.
- A fake delivery notice, written to make a courier look expected, is **refused by code reading the message itself**. A model may help read it. It decides nothing.

Measured, not demonstrated

Sixty courier visits, through an emulator that drops one webhook in ten.

60 / 60
families told
every household whose doorbell was seen

0
reminders played twice
duplicated deliveries are absorbed

54 / 60
reminders played
a lost doorbell press cannot be recovered

261
tests
and 48 rules broken on purpose

The misses are published next to the successes. Lose both doorbell webhooks for one visit and nobody is told — about one visit in a hundred, and the Partner API offers no way to ask the doorbell what it saw. A neighbour let in unannounced is a false alert.

What two days of walking the screens found

The suite proves the rules. Walking the screens proves the product.

Five defects, and **none of the 261 tests could see them** — they all call the API, and all five lived in the part a person touches.

- A required form field made **the one refusal this project exists for** unreachable: a collection message has no delivery window, so the browser never submitted it.
- A question rendered **below the fields it was meant to fill**. The first person to use it filled them by hand and never saw it.
- **“Call Margaret now” hid** behind the change alert a scammer’s next move produces a minute later.
- A report link in the alert e-mail needed a session **the phone it arrived on had never had**.

What it will not do

Recognise faces. Call the police.
Let a model decide. Run without her say.

It never tries to identify a visitor. False alerts exist, so people decide. Code checks every reading against the message itself. **She switches it on, chooses who is told, and one button deletes everything.**

Built on the Ring Partner API. Ring publishes no simulator, so every measurement above was taken against **ring-sandbox** (MIT), an open source emulator written by someone else — read in full, pinned by hash, and sent one pull request back.

github.com/Yoh5/callfirst

callfirst-bgtj.onrender.com/try

ring-sandbox · pull/10